INCH-POUND
MS27183K
16 June 2021
SUPERSEDING
MS27183J
11 March 1996

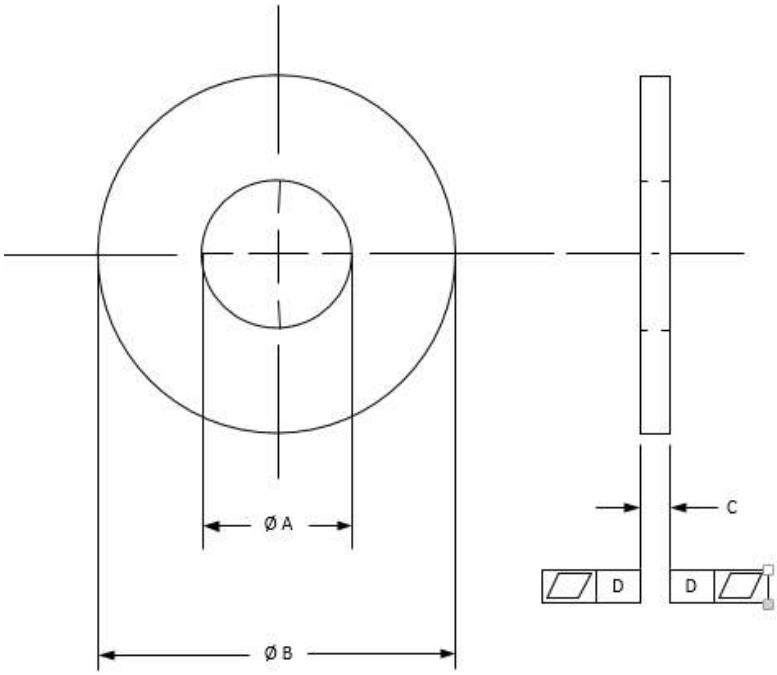
DETAILED SPECIFICATION SHEET

WASHER, FLAT (ROUND, STEEL, CADMIUM OR ZINC-NICKEL PLATED) GENERAL PURPOSE

This specification is approved for use by all components and agencies the Department Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following document listed in ASSIST specified in the solicitation FF-W-92.

Reactivated after 16 June 2021 and may be used for new and existing designs and acquisitions.



Comments, suggestions, or questions on this document should be addressed to U.S. Army CCDC, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-IES, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.ccdc-devcom.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

ASMC N/A

FSC: 5310

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TABLE I. Dash numbers and dimensions.

DASH NUMBER	A	TOLERANCE		B	TOLERANCE		C THICKNESS (BASIC)	TOLERANCE		D
		(+)	(-)		(+)	(-)		MAX	MIN	
-1	.078	.000	.005	.188	.000	.005	.020	.025	.016	.005
-2	.094	.000	.005	.250	.000	.005	.020	.025	.016	.005
-3	.125	.000	.005	.250	.000	.005	.022	.028	.017	.005
-4	.125	.008	.005	.312	.008	.005	.032	.040	.025	.005
-5	.156	.008	.005	.312	.008	.005	.035	.048	.027	.005
-43	.156	.008	.005	.250	.000	.015	.016	.020	.013	.005
-6	.156	.008	.005	.375	.015	.005	.049	.065	.036	.005
-7	.188	.008	.005	.375	.015	.005	.049	.065	.036	.005
-41	.188	.008	.005	.438	.015	.005	.049	.065	.036	.005
-8	.219	.008	.005	.438	.015	.005	.049	.065	.036	.005
-42	.219	.008	.005	.500	.015	.005	.049	.065	.036	.005
-44	.125	.008	.005	.375	.015	.005	.032	.040	.025	.005
-45	.141	.008	.005	.281	.005	.005	.032	.040	.025	.005
-46	.188	.008	.005	.500	.015	.005	.065	.080	.051	.005
-47	.219	.008	.005	.365	.005	.015	.032	.040	.025	.005
-48	.250	.005	.005	.020	.005	.005	.065	.040	.051	.005
-49	.250	.015	.015	.005	.015	.005	.032	.080	.025	.005
-50	.250	.015	.015	.005	.015	.005	.032	.040	.025	.005
-9	.250	.015	.015	.005	.015	.005	.065	.080	.051	.005
-10	281	.015	.015	.005	.015	.005	.065	.080	.051	.005
-51	.250	.015	.015	1.000	.015	.005	.065	.080	.051	.005
-52	.281	.015	.005	.734	.015	.007	.065	.080	.051	.005
-11	.312	.015	.005	.734	.015	.007	.065	.080	.051	.005
-12	.344	.015	.005	.688	.015	.007	.065	.080	.051	.005
-13	.375	.015	.005	.875	.030	.007	.083	.104	.064	.005
-53	.312	.015	.015	.875	.015	.015	.065	.075	.051	.005
-54	.312	.015	.005	.500	.015	.015	.035	.048	.027	.005
-55	.375	.015	.005	.734	.015	.015	.093	.094	.064	.005
-56	.375	.015	.005	.625	.030	.010	.035	.048	.027	.005
-57	.406	.015	.005	.875	.015	.007	.063	.070	.051	.005
-58	.412	.010	.005	.0625	.005	.005	.032	.040	.025	.005
-14	.406	.015	.005	.812	.015	.007	.065	.080	.051	.005
-15	.438	.015	.005	1.000	.030	.007	.083	.104	.064	.005
-59	.438	.015	.005	1.365	.015	.007	.083	.104	.064	.005
-60	.438	.015	.005	.875	.015	.007	.065	.080	.051	.005
-61	.438	.015	.005	1.325	.015	.007	.065	.104	.051	.005
-62	.469	.005	.020	.734	.030	.007	.065	.080	.051	.005
-16	.469	.015	.005	.992	.015	.007	.065	.080	.051	.005
-17	.500	.015	.005	1.250	.030	.007	.083	.104	.064	.005
-18	.531	.015	.005	1.062	.030	.007	.095	.121	.074	.005

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TABLE I. Dash numbers and dimensions - (Continued).

-19	.562	.015	.005	1.375	.030	.007	.109	.132	.086	.005
-20	.594	.015	.005	1.156	.030	.007	.095	.121	.074	.005
-63	.656	.015	.010	2.125	.030	.007	.095	.115	.080	.005
-21	.656	.030	.007	1.312	.030	.007	.095	.121	.074	.005
-22	.688	.030	.007	1.750	.030	.007	.134	.160	.108	.005
-23	.812	.030	.007	1.469	.030	.007	.134	.160	.108	.005
-24	.812	.030	.007	2.000	.030	.007	.148	.177	.122	.005
-25	.938	.030	.007	1.750	.030	.007	.134	.160	.108	.010
-26	.938	.030	.007	2.250	.030	.007	.165	.192	.136	.010
-27	1.062	.030	.007	2.000	.030	.007	.134	.160	.108	.010
-28	1.062	.030	.007	2.500	.030	.007	.165	.192	.136	.010
-29	1.250	.030	.007	2.750	.030	.007	.165	.192	.136	.010
-30	1.375	.030	.007	3.000	.030	.007	.165	.192	.136	.010
-31	1.500	.045	.010	3.250	.045	.010	.180	.213	.153	.010
-32	1.625	.045	.010	3.500	.045	.010	.180	.213	.153	.010
-33	1.750	.045	.010	3.750	.045	.010	.180	.213	.153	.010
-34	1.875	.045	.010	4.000	.045	.010	.180	.213	.153	.010
-35	2.000	.045	.010	4.250	.045	.010	.180	.213	.153	.010
-36	2.125	.045	.010	4.500	.045	.010	.180	.213	.153	.010
-37	2.375	.045	.010	4.720	.045	.010	.220	.248	.193	.010
-38	2.625	.045	.010	5.000	.045	.010	.238	.280	.210	.010
-39	2.875	.065	.010	5.250	.065	.010	.259	.310	.228	.010
-40	3.125	.065	.010	5.500	.065	.010	.284	.327	.249	.010

REQUIREMENTS

1. MATERIAL, STEEL, CARBON GRADES 1005 THRU 1020 (UNS G10050 THRU UNS G10200), IN ACCORDANCE WITH ASTM A108, ASTM A109 OR ASTM A569, ANY TEMPER.

2. PROTECTIVE COATING AND SURFACE TREATMENTS: CADMIUM PLATE /1 IN ACCORDANCE WITH SAE-AMS-QQ-P-416, TYPE II, CLASS 3 OR ZINC-NICKEL IN ACCORDANCE WITH MIL-PRF-32647, TYPE K16.

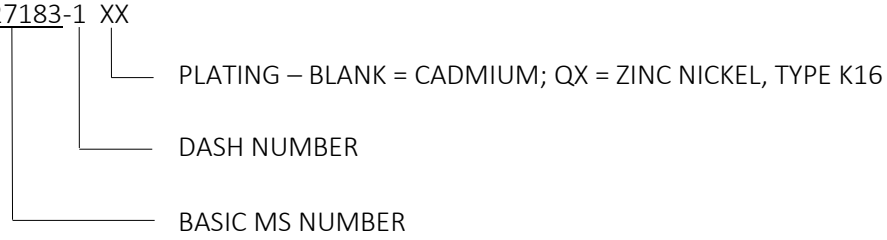
3. SURFACE TEXTURE, BEARING SURFACE ROUGHNESS SHALL NOT EXCEED 125 MICROINCHES (Ra) IN ACCORDANCE WITH ANSI/ASME B46.1.

/1 Cadmium is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 is the preferred method.

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4. PART NUMBER, THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I AND PLATING SUFFIX (IF REQUIRED).

EXAMPLE MS27183-1 XX



EXAMPLE PART NUMBERS:

MS27183-1 - WASHER, FLAT (ROUND STEEL), ID - .078, OD - .188, THICKNESS .020, CADMIUM PLATED.
MS27183-1QX - WASHER, FLAT (ROUND STEEL), ID - .078, OD - .188, THICKNESS .020, ZINC-NICKEL, TYPE K16 PLATED

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.

TABLE I. PART NUMBERS.

PART NUMBERS				
CANCELLED				PREFERRED
UNCOATED	PHOSPHATE	CAD OR ZN	CADMIUM	Zinc-Nickel Type K16
MS15795-1	MS15795-101	MS15795-201	MS27183-1	MS27183-1QX
-2	-102	-202	-2	-2QX
-3	-103	-203	-3	-3QX
-4	-104	-204	-4	-4QX
-5	-105	-205	-5	-5QX
-6	-106	-206	-6	-6QX
-7	-107	-207	-7	-7QX
-8	-108	-208	-8	-8QX
-9	-109	-209	-9	-9QX
-10	-110	-210	-10	-10QX
-11	-111	-211	-11	-11QX
-12	-112	-212	-12	-12QX
-13	-113	-213	-13	-13QX
-14	-114	-214	-14	-14QX
-15	-115	-216	-15	-15QX
-16	-116	-216	-16	-16QX
-17	-117	-217	-17	-17QX
-18	-118	-218	-18	-18QX
-19	-119	-219	-19	-19QX
-20	-120	-220	-21	-21QX
-21	-121	-221	-22	-22QX
-22	-122	-222	-23	-23QX
-23	-123	-223	-24	-24QX
-24	-124	-224	-25	-25QX
-25	-125	-225	-26	-26QX
-26	-126	-226	-27	-27QX
-27	-127	-227	-28	-28QX
-28	-128	-288	-29	-29QX
-29	-129	-229	-30	-30QX
-30	-130	-230	-31	-31QX
-31	-131	-231	-32	-32QX
-32	-132	-232	-33	-33QX
-33	-133	-233	-34	-34QX
-34	-134	-234	-35	-35QX
-35	-135	-235	-36	-36QX
-36	-136	-236	-37	-37QX
-37	-137	-237	-38	-38QX
-38	-138	-238	-39	-39QX
-39	-139	-229	-40	-40QX
-40	-140		-20	-20QX

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2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS DOCUMENT AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS DOCUMENT SHALL TAKE PRECEDENCE.
3. UNLESS OTHERWISE SPECIFIED, ISSUE OF REFERENCED DOCUMENTS ARE THOSE IN EFFECT AT THE TIME OF SOLICITATION.
4. CADMIUM OR ZINC COATED STEEL WASHERS ONLY, PREVIOUSLY SPECIFIED ON MS15795 WERE CANCELLED AND SUPERSEDED BY CADMIUM, ZINC, OR ZINC-NICKEL PLATED STEEL WASHERS SPECIFIED HEREIN
5. UNCOATED STEEL WASHERS AND PHOSPHATE COATED STEEL WASHERS, COVERED BY DASH NUMBERS GIVEN ON MS15795, ARE CANCELLED/INACTIVATED AFTER THE INDICATED APPROVAL
6. USE THE DASH NUMBERS GIVEN IN TABLE II
7. IN THE PRECEDING SHEET OF THIS STANDARD, THE CANCELLED WASHERS CAN NOT ALWAYS REPLACE
8. EXHISTING STOCKS OF WASHERS SHOULD BE USED UNTIL EXISTING STOCK IS DEPLETED
9. USE ZINC-NICKEL WASHERS FOR NEW DESIGN AND REPLACEMENT
10. THE MARGINS OF THIS SPECIFICATION ARE MARKED WITH VERTICAL LINES TO INDICATE WHERE CHANGES FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE PREVIOUS ISSUE.

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